

|                                                                                     |                                                                           |                                                                     |                                                                         |                                                                                       |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <div>+90%</div> <div>CUMULATIVE RETURN</div> <div>XLU (buy &amp; hold): +113%</div> | <div>8.1%</div> <div>ANN. RETURN</div> <div>XLU ann. return: +12.0%</div> | <div>0.31</div> <div>SHARPE RATIO</div> <div>XLU Sharpe: 0.24</div> | <div>-13.7%</div> <div>MAX DRAWDOWN</div> <div>XLU Max DD: -36.1%</div> | <div>0.15</div> <div>MEAN IC (ALL MONTHS)</div> <div>Positive in ~58% of months</div> |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------|

THE THESIS

Utilities with genuine grid resilience — revealed by how their physical assets fare during power market stress — earn a structural premium over time; less resilient operators suffer a persistent discount.

Some of this is eventually captured in earnings, but stress beta derived from physical topology may identify winners and losers before it shows up in reported financials.

This strategy exploits that lead time: ranking utilities by their live stress beta to go long the most resilient operators and short the most exposed.

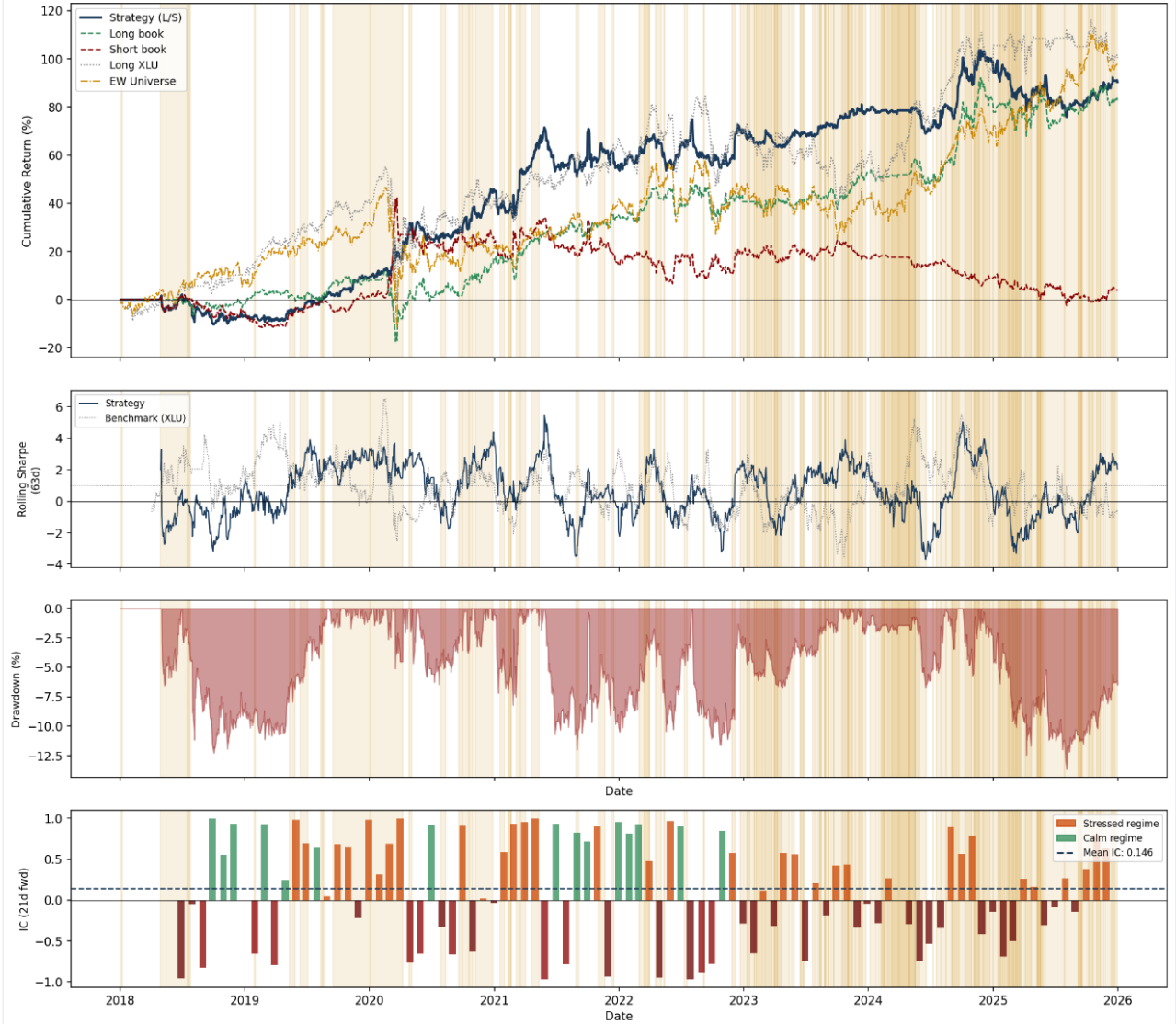
In doing so, the strategy directs capital toward operators investing in grid stability and withdraws it from those who are not — a real mechanism for rewarding responsible grid investment.

On raw vs. risk-adjusted returns: XLU buy-and-hold returned +113% over the same period versus the strategy's +90%, but with a max drawdown of ~36.1% (vs ~13.7%) and annual volatility of 20.6% (vs 13.2%) — the strategy delivers a comparable return per unit of risk with dramatically less downside exposure.

The sharpest exception was January 27, 2025: the strategy was overweight VST, on the thesis that surging data center power demand would reward resilient generators — a sound fundamental call that was overwhelmed by DeepSeek's claim that large AI models could run at a fraction of the energy cost.

Grid Resilience Strategy — Backtest Performance

Sharpe: 0.319 | Ann. Ret: 8.1% | Max DD: -13.7% | Bench Sharpe: 0.235



Top: Cumulative return vs. XLU and equal-weight utility universe. Orange shading = stressed grid regime. Second: Rolling 63-day Sharpe vs. XLU benchmark. Third: Strategy drawdown — peak-to-trough never exceeds ~13.7%, versus ~36.1% for XLU. Bottom: Monthly Information Coefficient. Orange bars = stressed months. Mean IC 0.15 suggests predictive power of the factor for returns.

HOW THE SIGNAL IS BUILT

| Step | What                                        | Source                                                                                                                                |
|------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Daily LMP prices across 5 ISOs              | gridstatus                                                                                                                            |
| 2    | Grid Stress Index (GSI) per ISO             | Composite 4-signal score: Energy Price (40%), Congestion Fraction (25%), Reserve tightness (20%), Weather and Congestion Events (15%) |
| 3    | Conditional stress beta per stock           |                                                                                                                                       |
| 4    | Use stress beta to create portfolio weights |                                                                                                                                       |

ISSUES TO BE HANDLED

- Data depth** As this is still a sketch the data depth is low. Also the strategy stats do not consider slippage, transactions/rebalancing costs, etc.
- Data coverage** SPP and CAISO LMP history is shorter than ERCOT/PJM, making stress betas for those utilities noisier.
- Redundancy check** GSI beta must perform orthogonally to standard quality factors (ROE, leverage, dividend stability) to ensure that this GSI index is not just another version of already-traded quality (Axioma style factors will help)
- Improvement** Addition of Interest Coverage Ratio (ICR) refines the GSI by helping to identify financially fragile utilities entering stress events.
- Improvement** Inter-zonal congestion spread — LMP component decomposition (energy / congestion / loss) to sharpen the congestion sub-signal beyond the current fraction proxy.
- Improvement** Addition of AI/DataCenter Sentiment Power Futures, Equity Short Interest (and, if sufficiently liquid, options signals) may help to distinguish grid-fundamental moves from macro narrative shocks that can temporarily reprice even the most resilient names.